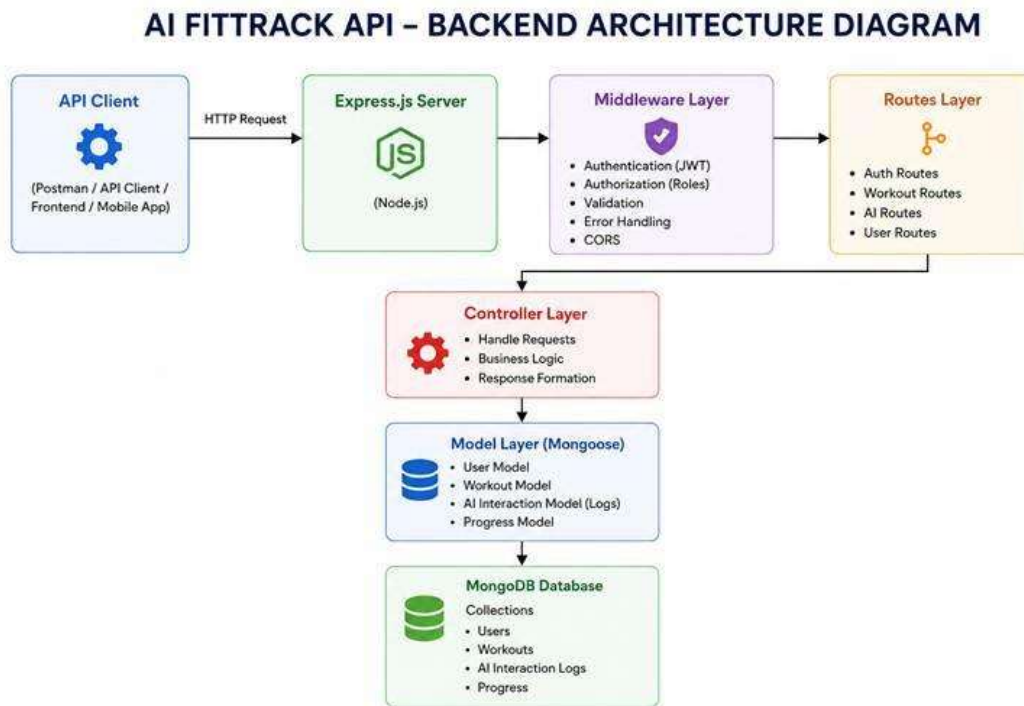


Phase 3

Project Design Phase Phase

Phase 3: Project Design Phase



The FitTrack AI backend uses a layered Model-View-Controller (MVC) architecture that separates routing, middleware, business logic, database operations, and AI services into independent modules with single responsibilities. Incoming HTTP requests to the Express server are directed to specific routes, passing through JWT authentication middleware on protected endpoints before reaching the controllers. The controllers then coordinate with Mongoose models for data persistence and a dedicated Google Gemini AI service layer to process requests and return structured JSON responses, ensuring high security, maintainability, and scalability.

Overall Architecture Components

- **Client & Routing Layer:** Client Application (React, Mobile App, Postman), Express Server, API Routes, and Authentication Middleware.
- **Logic & AI Layer:** Controllers, Services, and Google Gemini AI Service.
- **Data Layer:** Mongoose Models and MongoDB Database.

All components communicate through well-defined interfaces, ensuring loose coupling and easier future enhancements.